

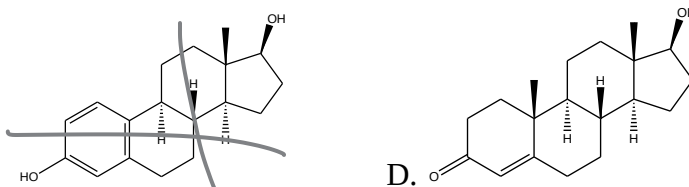
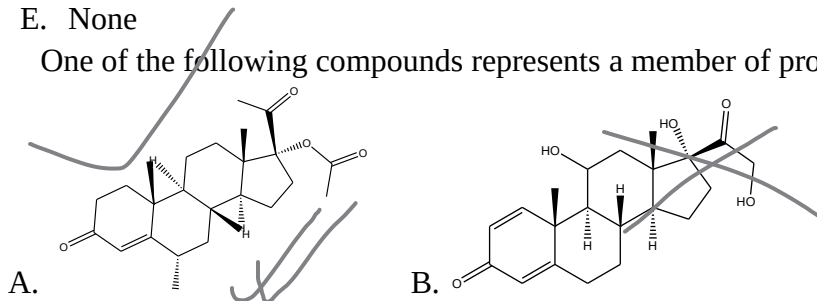
Addis Ababa University
College of Health Science, School of Pharmacy
Department of Pharmaceutical Chemistry and Pharmacognocny
Medicinal Chemistry I Assignment

Name: ----- ID. No. ----- Section -----

I. Choose the best Answer

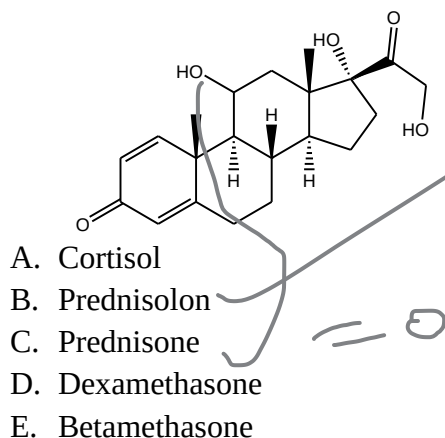
1. Estrogens are clinically applied in all of the following conditions except
 - A. In case of ovarian failure
 - B. To relief menopausal symptoms
 - C. In case of dysmenorrheal
 - D. As anabolic agents
 - E. None
2. One is different structurally
 - A. Estradiol
 - B. Mestranol
 - C. Diethylstilbesterol
 - D. Estrone
 - E. None
3. Which is **not** true about the chemistry of polypeptide hormones?
 - A. Their biosynthesis involves dehydration reaction
 - B. The individual amino acids are linked to each other by amide bond
 - C. All contain steroidal nucleus
 - D. Are biosynthesized from L-amino acids
4. One does not refer to insulin hormone.
 - A. Chain A is acidic in nature
 - B. Consists of only two disulfide bonds three
 - C. Chain B is basic in nature
 - D. The peptide chains are connected hydrophobic, hydrogen bonds and salt bridge in addition to the disulfide bonds
 - E. None
5. Choose the correct statement about structural activity relationship of insulin
 - A. All amino acids of chain B are important for its activity
 - B. All amino acids of chain A are important for its activity
 - C. All disulfide bonds are important for insulin activity
 - D. Replacement of amino acids of chain A with other L-amino acids does not have significant effect on insulin activity
 - E. All except A

6. All are true about the natural estrogens except
- Ring A is benzoid
 - All have phenolic OH at C-3 as β position
 - They lack a carbon side chain at C-17 position
 - CH₃ at C-10 is absent
 - None
7. One antagonizes the actions of estrogens
- Mestranol
 - Estradiol
 - Tamoxiphen
 - Ethinylestradiol
 - None
8. One of the following compounds represents a member of progestins.

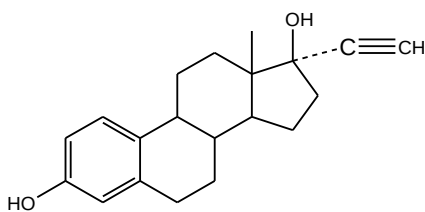


9. One refers to the functions of glucocorticoids
- Maintaining electrolyte and water balance
 - As antiinflammatory agent
 - As immunesuppressants
 - Maintaining the brain function

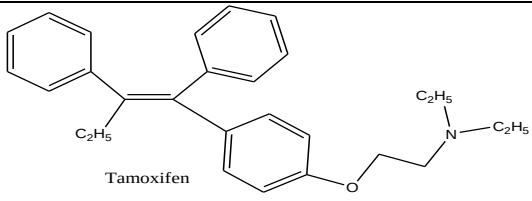
10. The following chemical structure represents one of the following corticosteroid drugs



11. One of the following drugs serves as antiandrogen agent
- A. Mifepristone
 - B. Estrogens
 - C. Methyltestosterone
 - D. Medroxyprogesterone
 - E. None
12. One of the following is **not true** regarding cortical steroids
- A. Oxygen at C-11 and 17α -OH enhances glucocorticoid activity
 - B. Introduction of fluoro group at 6α or 17α enhances both mineralo & glucocorticoid activities
 - C. Insertion of an alkene link at C-1- C-2 enhances glucocorticoid & anti-inflammatory properties & reduce mineralocorticoid activity
 - ☒ D. Increased mineralocorticoid activity reduces risk of Na retention & edema

13. 	I. It is a steroidal estrogen II. It is a non-steroidal estrogen III. It is ethinylestradiol IV. Equally potent as estradiol by injection but 15-20X more active orally
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- A. I only
- B. II only
- C. I, II and III
- D. I and III
- E. I, III and IV

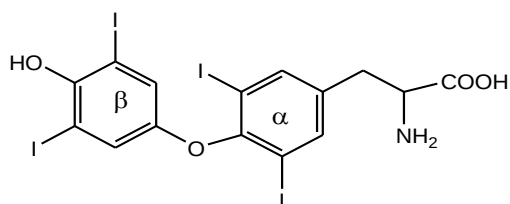
14.  Tamoxifen	I. It is a steroidal anti-estrogen II. It is a non-steroidal anti-estrogen III. Used for treatment of breast cancer
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- A. I only
- B. II only

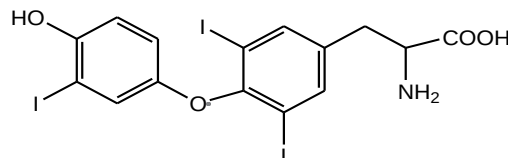
C. I, II and III

D. II and III

15. One of the following clearly describe about thyroxine and Liothyronine



Thyroxine / tetraiodothyronine / T₄



Liothyronine / triiodothyronine / T₃

A. aromatic rings should be parallel to each other

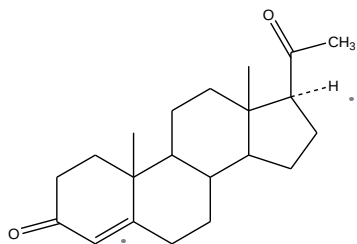
B. Replacement of oxygen by sulfur reduce activity

☒ C. Replacement 4' hydroxy by methyl group gives an inactive analog

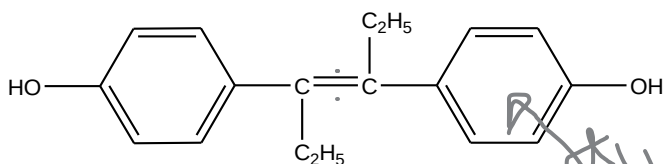
D. Reversed T3 is active

II. Answer the following questions briefly in the space provided

1. The following drug is not active orally and has short duration action explain the reason and suggest possible solutions to overcome those problems.



2. Summarize structure activity relationship of diethylstilbesterol (DES)



3. Insulin is a polypeptide hormone consisting of two chains (Chain A & Chain B) in its structure. Number of amino acids in chain A is 21 & in chain B is 30
_____ Of the two chains A is acidic & B is basic in character.

